Table S4. Additional module validation panels

Function	Independent reference	Observed result	Scope limit
Promoter extraction	Direct 1-based inclusive FASTA slicing and reverse complement; bedtools 2.31.1	8/8 exact sequences, 2,200 bp each, both strands represented	Eight loci on one Arabidopsis chromosome
CDS sequence quality	Independent base-count/frame/start/stop oracle	Four RBCS CDS models (411, 561, 546, 546 nt) matched; no ambiguous bases flagged	Four models from one family
Protein similarity	Test-side Smith–Waterman and BLASTP 2.12.0+	All 16 within-panel alignment metric checks; 12/12 non-self RBCS and 8/8 negative pair classifications agree; 20/20 in the cross-species mature-RBCS panel	Small one-family smoke test; not orthology, sensitivity, or specificity evaluation
Expression grouping	Expression Atlas E-MTAB-7978 source values	Selected means/medians reproduced; measured zeros retained	Single-assay group arithmetic, not differential expression
Domain mapping	Independent Python reconstruction	16 domain/deletion coordinate records across 13 genes matched	Existing annotated models; not a curated multi-species accuracy set
Project persistence	JavaFX regression actions	Save, replace, restore, locus state and outputs verified in CI	Software behavior, not usability study